

(Software for) Modeling and Solving Multiobjective Optimization Problems

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Nantes Université, France

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Organization

Part 1 (9:00-10:30):

Fundamentals

Building a complete workflow: from modeling to solving and analyzing

Part 2 (11:00-12:30):

Case Studies

Applying the workflow to increasingly challenging case studies

Slides, examples, notebooks

Resources online:

https://github.com/xgandibleux/EURO_MCDM2026/

Legend of the marks on the slides:



Additional resources provided



Live demonstration



Instruction(s) to be written

Preamble

Why this adventure ?

More than 30 years of research into multi-objective optimisation

$$\min \left\{ Fx : x \in X \right\}$$

2015: Kick-off of the ANR/DFG research project **vOpt**

Sub-task 2.2: Experimental Analysis and Prototype Development.

- ▶ A consortium composed of mathematicians and informaticians
- ▶ MOMILP: Multi-Objective Mixed Integer Linear Programming

$$\min \left\{ Cx + Dy : Ax + By = d, x \geq 0, y \in \mathbb{Z} \right\}$$

↳ The project of a **multi-objective MILP solver** was born!

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Purposes of the solver

Usages:

Natural, intuitive use for mathematicians, informaticians, engineers

- ▶ for research:
support and primitives for the development of new algorithms
- ▶ for experimentation:
methods and algorithms for performing numerical experiments
- ▶ for teaching:
environment for practicing of theories and algorithms

Expectations:

Easy to formulate a problem, provide the data, solve a problem, collect the outputs, analyze the solutions...

- + Solver: robust, efficient, evolutive
- + Software: free, open source, multi-platform

...and easy to install: no need of being expert in computer science!

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- ▶ Matlab, Scilab or Mathematica
- ▶
- ▶



Raise you hand if you are familiar with

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- ▶ Python, or R
- ▶



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- ▶ Python, or R
- ▶ Fortran, C, or C++



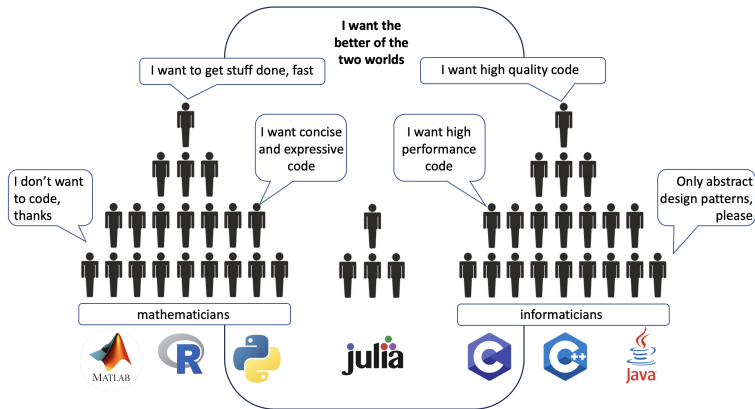
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Maths formalisms as Matlab
Interactive as Python
Fast as C/C++

Why Julia and JuMP for MOO?



- ↳ **Julia** progr. language: easy, high-level, efficient, expressive
- JuMP** modeling language: expressive, efficient, evolutive

High-level and expressive?

C

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int A[100][100][100];
    int i,j,k,loop;
    for (loop=0; loop <1000; loop++){
        for (i=0; i<100; i++){
            for (j=0; j<100; j++){
                for (k=0; k<100; k++){
                    A[i][j][k] = loop;
                }
            }
        }
        printf("%d\n", A[99][99][99]);
    }
    return 0;
}
```

Julia

```
@time begin
    A = zeros(100,100,100)
    for loop = 1:1000
        for i = 1:100
            for j = 1:100
                for k = 1:100
                    A[i,j,k] = loop
                end
            end
        end
        println(A[100,100,100])
    end
end
```

MATLAB

```
tic
    A = zeros(100,100,100);
    for loop = 1:1000
        for i = 1:100
            for j = 1:100
                for k = 1:100
                    A(i,j,k) = loop;
                end
            end
        end
        disp(A(100,100,100));
    end
toc
```



Efficient?

► C

without optimization option

1.286582 seconds

with optimization option -O3

0.087682 seconds

► Julia

1-to-1 straightforward implementation

26.732819 seconds (489.05 M allocations: 7.297 GiB, 0.29% gc time)

julia style compliant implementation

0.252995 seconds (22.39 k allocations: 8.499 MiB)

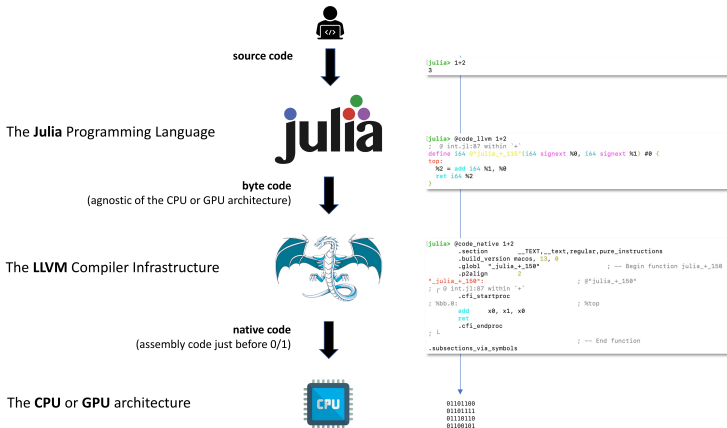
polished implementation

0.114872 seconds (20.75 k allocations: 4.318 MiB)



Efficient! Why?

LLVM: middleman between source code and compiled native code



Easy to use?



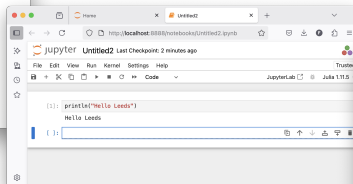
...the Julia REPL

```
xavierg — Julia — julia — 80x24
Last login: Fri May 30 13:26:56 on ttys003
(base) xavierg@Xaviers-iMac ~ % julia

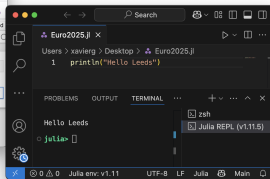
Documentation: https://docs.julialang.org
Type ** for help, **j* for pkg help.
Version 1.11.5 (2025-04-14)
Official https://julialang.org/ release

julia> println("Hello Leeds")
Hello Leeds
julia> 
```

...a notebook (e.g. jupyter)



...an IDE (e.g. VScode)



... or also in command line into a terminal.



Julia in a nutshell...

Main key concepts:

- ▶ free, open-source, multi-platform
- ▶ Julia has an LLVM-based Just-in-time compilation (JIT)
- ▶ Compilation down to native code (running on CPU and GPU, ...)
- ▶ Julia is dynamically typed, provides multiple dispatch
- ▶ Julia has many built-in mathematical functions
- ▶ A built-in package manager
- ▶ Designed for parallelism and distributed computation

Resources:

The Julia Programming Language (<https://julialang.org/>):
download, documentation, resources, community, etc.

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Multi-Objective Mathematical Programming Software

Multiobjective Optimization Problem (MOP) notations

$$\begin{array}{ll}\min & F(x) \\ \text{s.t.} & x \in X\end{array}$$

where:

$$\begin{array}{ll}X \subseteq \mathbb{R}^n & \longrightarrow \text{the set of feasible solutions} \\ Y = F(X) \subseteq \mathbb{R}^p & \longrightarrow \text{the set of images}\end{array}$$

Computing sets X_E and Y_N for MOP:

$$y = F(x)$$

$y^* \in Y$ is nondominated, if $\nexists y \in Y$ such that $y_i \leq y_i^*, \forall i$ and $y \neq y^*$.

Y_N is the set of nondominated points.

$x^* \in X$ is efficient if y^* is nondominated.

X_E is a complete set of efficient solutions.

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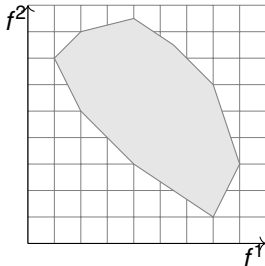
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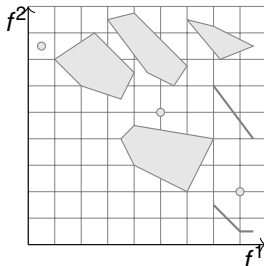
Examples of Y and Y_N when $p = 2$

Computing Y_N for Multiobjective (linear) Optimization Problems (MOP):

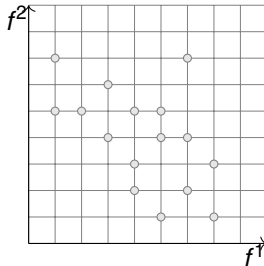
$$x \in \mathbb{R}^{n_1}$$



$$x \in \mathbb{R}^{n_1} \times \mathbb{Z}^{n_2}$$



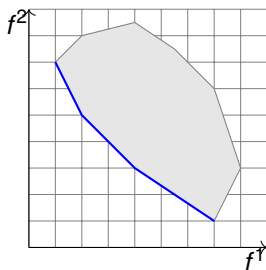
$$x \in \mathbb{Z}^{n_2}$$



Examples of Y and Y_N when $p = 2$

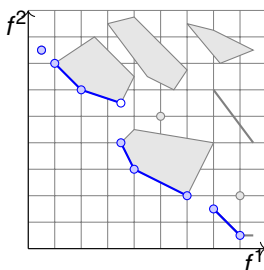
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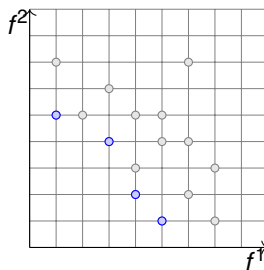
Continuous
nondominated set
(edges)

$$x \in \mathbb{R}^{n_1} \times \mathbb{Z}^{n_2}$$



Nondominated set
composed of edges
that are either closed,
half-open, open or
reduced to a point

$$x \in \mathbb{Z}^{n_2}$$

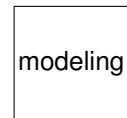


Discrete set of
nondominated
points

Multi-Objective Mathematical Programming Software

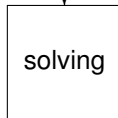
- JuMP
- GNU MP
- Pyomo...

- AMPL
- GAMS
- OPL...



low-level formulation
(flat form)

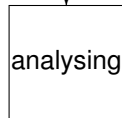
- MPS
- LP...



- HiGHS
- GLPK
- SCIP...

- GUROBI
- CPLEX
- Xpress...

high-level model
(structured form)



- ...

Software for modeling activity

Modelers: software tools devoted to express mathematical optimisation problems

- ▶ Algebraic Modeling Language (AML):

a high-level programming language designed for mathematical optimisation with an emphasis on declarative modeling and independent of any particular solver.

- ▶ expressed in proprietary languages:
GAMS, AMPL, AIMMS, OPL, Mosel, GMP, ZIMPL, ...
- ▶ close to programming languages:
JuMP (Julia), Pyomo (Python),...

- ▶ Solver-Oriented Interface (SOI):

tightly coupled to a specific optimisation solver; provide programmatic control and access to solver-specific features within a host programming language (C/C++, Java, Python,...)

- ▶ Application Programming Interfaces (API):
API C++ Gurobi,...
- ▶ embedded Domain-Specific Languages (DSL) :
highspy, MATLAB, Hexaly Modeler,...

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Software for solving activity

Optimisation solver: a specialized software engine designed to find solutions to mathematical programming problems

- ▶ Generic MILP solvers:

implement numerical algorithms tailored to problem classes (LP, MIP,...)

- ▶ Gurobi, CPLEX, Xpress, MOSEK, Hexaly, COPT,...
- ▶ GLPK, SCIP, HiGHS, CBC/CLP,...

- ▶ Specific MOMP solvers:

specialized optimisation engines designed to directly handle problems with multiple objectives

- ▶ ADBASE, BENSOLVE, INNER, PaMILO, PolySCIP, vOptSpecific

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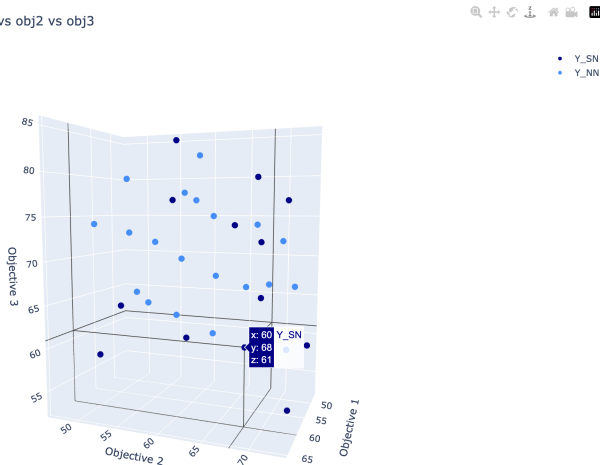
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Software for analysing activity

Homemade...

Objective space : obj1 vs obj2 vs obj3

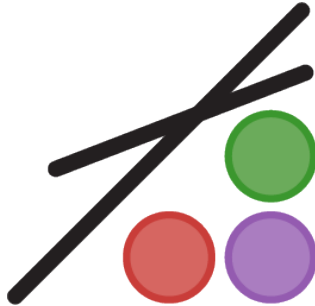


Summary of software examined



<i>Software</i>	<i>Commercial</i>	<i>Open-source</i>	<i>Modeler</i>	<i>Solver</i>	<i># Trade-offs</i>
MATLAB	✓	–	SOI	–	≥1
Gurobi Optimizer	✓	–	SOI	generic	1
HiGHS	–	✓	SOI	generic	1
COPT	✓	–	SOI	generic	1
Hexaly	✓	–	SOI	generic	1
FICO Xpress Optimizer	✓	–	AML+SOI	generic	1
CPLEX Optimisation Studio	✓	–	AML+SOI	generic	1
AMPL	✓	–	AML	–	1
AIMMS	✓	–	AML	–	1
GAMS	✓	–	AML	–	≥1
Pyomo	–	✓	AML	–	–
PuLP	–	✓	AML	–	–
CVXPY	–	✓	AML	specific	1
MOCVXPY	–	✓	AML	specific	≥1
BENSOLVE	–	✓	–	specific	≥1
INNER	–	✓	–	specific	≥1
PaMILO	–	✓	–	specific	≥1
SCIP with ZIMLP and PolySCIP	–	✓	AML	specific	≥1
JuMP with MultiObjectiveAlgorithms	–	✓	AML	meta-solver	≥1

JuMP



version 1.30.0 and more

What is JuMP?

An Algebraic Modeling Language (AML) among the existing ones . . .



... for mathematical optimization (linear, mixed-integer, conic, semidefinite, nonlinear) written in the Julia language.

Overview of JuMP

- ▶ **User friendliness:** syntax that mimics natural mathematical expressions.
- ▶ **Speed:** similar speeds to special-purpose modeling languages such as AMPL.
- ▶ **Solver independence:** Currently 56 solvers are supported.
Among the (M)LP solvers: GLPK, Cbc, Clp, HiGHS, SCIP, CPLEX, Gurobi, FICO Xpress, COPT...
- ▶ **Ease of embedding:** JuMP itself is written purely in Julia.
Solvers are the only binary dependencies.

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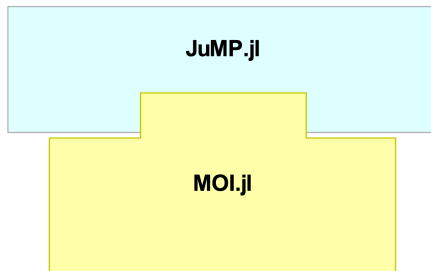
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Getting started with JuMP



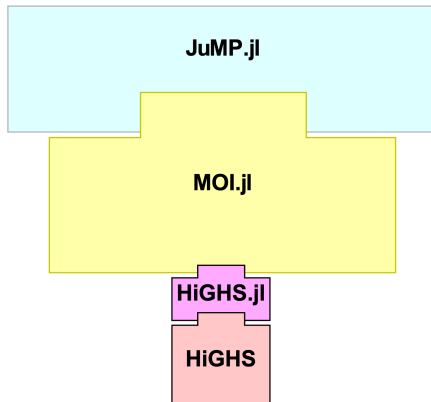
```
julia> using Pkg  
julia> Pkg.add("JuMP")
```



JuMP uses *MathOptInterface* (MOI), an abstraction layer designed to provide a unified interface to mathematical optimization solvers.

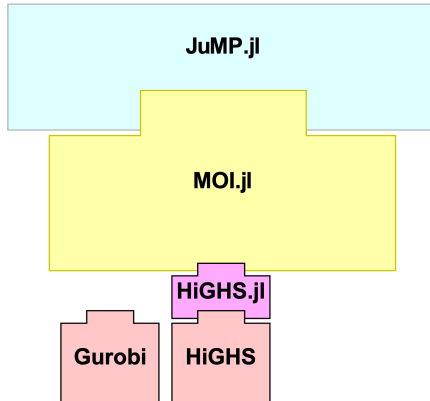


```
julia> Pkg.add("HiGHS")
```



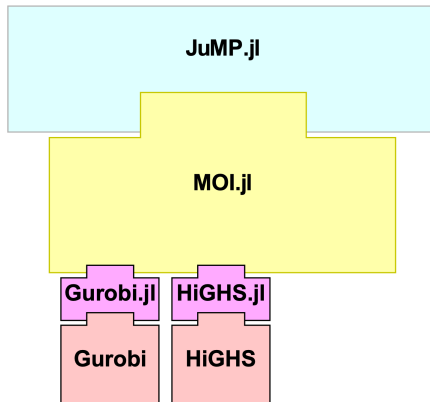
Getting started

With Gurobi or CPLEX or Xpress MP solver properly installed on your computer...



Getting started

```
julia> Pkg.add("Gurobi")
```



Explicit Modeling and Solving of an (MI)LP with JuMP

The optimization problem

$$\left[\begin{array}{rcll} \max z(x) = & x_1 & + & 3x_2 & (0) \\ s.t & x_1 & + & x_2 & \leq 14 & (1) \\ & -2x_1 & + & 3x_2 & \leq 12 & (2) \\ & 2x_1 & - & x_2 & \leq 12 & (3) \\ & x_1 & , & x_2 & \geq 0 & (4) \end{array} \right]$$

Declaring the packages to use



```
julia> using JuMP, HiGHS
```



Creating a model:

```
modelId = Model( )
```

```
julia> modLP = Model( )
```

Result:

[]



Defining a variable:

```
@variable(modelId, variable)
```

```
julia> @variable(modLP, x1 >= 0)
```

```
julia> @variable(modLP, x2 >= 0)
```

Result:

$$\left[\begin{array}{c} x_1, \quad x_2 \geq 0 \quad (4) \end{array} \right]$$



Defining an objective function:

```
@objective(modelId, sense, expression)
```

```
julia> @objective(modLP, Max, x1 + 3x2)
```

Result:

$$\left[\begin{array}{rcll} \max z(x) = & x_1 & + & 3x_2 & (0) \\ & x_1 & , & x_2 & \geq 0 & (4) \end{array} \right]$$



Defining a constraint:

```
@constraint(modelId, id, constraint)
```

```
julia> @constraint(modLP, cst1, x1 + x2 <= 14)
julia> @constraint(modLP, cst2, -2x1 + 3x2 <= 12)
julia> @constraint(modLP, cst3, 2x1 - x2 <= 12)
```

Result:

$$\left[\begin{array}{rcllcl} \max z(x) = & x_1 & + & 3x_2 & & (0) \\ s.t & x_1 & + & x_2 & \leq & 14 & (1) \\ & -2x_1 & + & 3x_2 & \leq & 12 & (2) \\ & 2x_1 & - & x_2 & \leq & 12 & (3) \\ & x_1 & , & x_2 & \geq & 0 & (4) \end{array} \right]$$

Solving the optimization problem



```
set_optimizer(modelId, solver )
```

```
julia> set_optimizer(modLP, HiGHS.Optimizer)
```

```
set_silent(modelId )
```

```
julia> set_silent(modLP)
```

```
optimize!(modelId)
```

```
julia> optimize!(modLP)
```

Solving the optimization problem



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```
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```
julia> set_silent(modLP)
```

```
optimize!(modelId)
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```
julia> optimize!(modLP)
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```
optimize!(modelId)
```

```
julia> optimize!(modLP)
```



```
julia> @show is_solved_and_feasible(modLP)
```

```
julia> @show termination_status(modLP)
```

```
julia> @show objective_value(modLP)
```

```
julia> @show value(x1)
```

```
julia> @show value(x2)
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```

Summary (explicit model)



```
julia> using JuMP, HiGHS

julia> modLP = Model( )
julia> @variable(modLP, x1 >= 0)
julia> @variable(modLP, x2 >= 0)
julia> @objective(modLP, Max, x1 + 3x2)
julia> @constraint(modLP, cst1, x1 + x2 <= 14)
julia> @constraint(modLP, cst2, -2x1 + 3x2 <= 12)
julia> @constraint(modLP, cst3, 2x1 - x2 <= 12)

julia> set_optimizer(modLP, HiGHS.Optimizer)
julia> set_silent(modLP)
julia> optimize!(modLP)

julia> @show is_solved_and_feasible(modLP)
julia> @show objective_value(modLP)
julia> @show value(x1), value(x2)
```

↳ https://github.com/xgandibleux/EURO_MCDM2026/blob/main/3.examples/1LPexplicit.ipynb

Details on

Variables:

- ▶ by default, the variables are **continuous** and **unbounded**:

```
julia> @variable(model, x)           # x is free
```

- ▶ possible to setup lower and/or upper bounds on a variable:

```
julia> @variable(model, x ≥ lb))    # x is bounded  
julia> @variable(model, x ≤ ub)  
julia> @variable(model, lb ≤ x ≤ ub)  
julia> @variable(model, x == 2)     # x is fixed
```

- ▶ possible to specify the type of a variable:

```
julia> @variable(model, x ≥ 0, Int)    #  $x \in \mathbb{N}$   
julia> @variable(model, x, Bin)       #  $x \in \{0, 1\}$ 
```

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```

Implicit Modeling and Solving of an (MI)LP with JuMP

Writing an implicit model (1/5)

Example (cont'd):

$$\left[\begin{array}{rcll} \max f(x) = & x_1 & + & 3x_2 & (0) \\ \text{s.t} & x_1 & + & x_2 & \leq 14 \quad (1) \\ & -2x_1 & + & 3x_2 & \leq 12 \quad (2) \\ & 2x_1 & - & x_2 & \leq 12 \quad (3) \\ & x_1 & , & x_2 & \geq 0 \quad (4) \end{array} \right]$$

↓

$$c = (1, 3), \quad d = (14, 12, 12), \quad T = \begin{pmatrix} 1 & 1 \\ -2 & 3 \\ 2 & -1 \end{pmatrix}$$

$$\left[\begin{array}{rcll} \max f(x) = & \sum_{j=1}^2 c_j x_j & & (0) \\ \text{s.t} & \sum_{j=1}^2 t_{ij} x_j \leq d_i & i = 1, 3 & (1 - 3) \\ & x_j \geq 0 & j = 1, 2 & (4) \end{array} \right]$$



Example (cont'd):

$$c = (1, 3), \quad d = (14, 12, 12), \quad T = \begin{pmatrix} 1 & 1 \\ -2 & 3 \\ 2 & -1 \end{pmatrix}$$

```
julia> c = [1, 3]
julia> d = [14, 12, 12]
julia> T = [1 1 ; -2 3; 2 -1]
julia> n,m = size(T)
```



Example (cont'd):

$$\left[\begin{array}{ll} \max z(x) = & \sum_{j=1}^m c_j x_j & (0) \\ \text{s.t} & \sum_{j=1}^m t_{ij} x_j \leq d_i & i = 1, n \quad (1-3) \\ & x_j \geq 0 & j = 1, m \quad (4) \end{array} \right]$$

```
julia> using JuMP, HiGHS
julia> md = Model(HiGHS.Optimizer)
```

```
julia> @variable(md, x[1:m] ≥ 0)
```

```
julia> @objective(md, Max, sum(c[j]*x[j] for j=1:m))
```

```
julia> @constraint(md, cst[i=1:n], sum(T[i,j]*x[j]
                                     for j=1:m) ≤ d[i])
```



Example (cont'd):

$$\left[\begin{array}{ll} \max z(x) = & \sum_{j=1}^m c_j x_j & (0) \\ \text{s.t} & \sum_{j=1}^m t_{ij} x_j \leq d_i & i = 1, n \quad (1-3) \\ & x_j \geq 0 & j = 1, m \quad (4) \end{array} \right]$$

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```



Example (cont'd):

$$\left[\begin{array}{ll} \max z(x) = & \sum_{j=1}^m c_j x_j & (0) \\ \text{s.t.} & \sum_{j=1}^m t_{ij} x_j \leq d_i & i = 1, n \quad (1-3) \\ & x_j \geq 0 & j = 1, m \quad (4) \end{array} \right]$$

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Example (cont'd):

$$\left[\begin{array}{ll} \max z(x) = & \sum_{j=1}^m c_j x_j & (0) \\ \text{s.t} & \sum_{j=1}^m t_{ij} x_j \leq d_i & i = 1, n \quad (1-3) \\ & x_j \geq 0 & j = 1, m \quad (4) \end{array} \right]$$

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```

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julia> md = Model(HiGHS.Optimizer)
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julia> @variable(md, x[1:m] ≥ 0)
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julia> @objective(md, Max, sum(c[j]*x[j] for j=1:m))
```

```
julia> @constraint(md, cst[i=1:n], sum(T[i,j]*x[j]  
                                     for j=1:m) ≤ d[i])
```

Structured variables in JuMP

- ▶ one-based integer ranges...

```
julia> @variable(md, x[1:4] ≥ 0)
```

```
julia> @variable(md, x[1:2, 1:2] ≥ 0, Int)
```

- ▶ ...but not necessarily...

```
julia> @variable(md, x[-4:4] ≥ 0)
```

- ▶ ...nor necessarily numerical

```
julia> @variable(md, x[:, :tram, :train], Bin)
```

More on solutions

Primal solution values for a scalar variable:

```
value(variable)
```

```
julia> value(x1)
```

Primal solution values for a structured variable:

```
value(variable[index])
```

```
julia> value(x[2])
```

```
value.(variable)
```

```
julia> value.(x)
```

More on solutions

Primal solution values for a scalar variable:

```
value(variable)
```

```
julia> value(x1)
```

Primal solution values for a structured variable:

```
value(variable [index])
```

```
julia> value(x[2])
```

```
value. (variable)
```

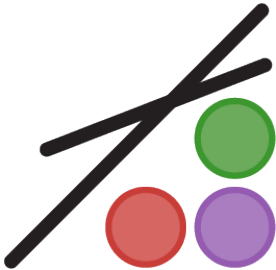
```
julia> value. (x)
```

Summary (implicit model)



↳ https://github.com/xgandibleux/EURO_MCDM2026/blob/main/3.examples/1LPimplicit.ipynb

JuMP and MultiObjectiveAlgorithms



version 1.30.0 and more



version 1.12.0 and more

The MOO problem

Compute X_E and Y_N for the following MOO problem:

$$\begin{array}{llllll} \max z_1 & = & x_1 & + & x_2 & \\ \min z_2 & = & x_1 & + & 3x_2 & \\ s.t. & & 2x_1 & + & 3x_2 & \leq 30 \\ & & 3x_1 & + & 2x_2 & \leq 30 \\ & & x_1 & - & x_2 & \leq 5.5 \\ & & x_1 & , & x_2 & \in \mathbb{N} \end{array}$$



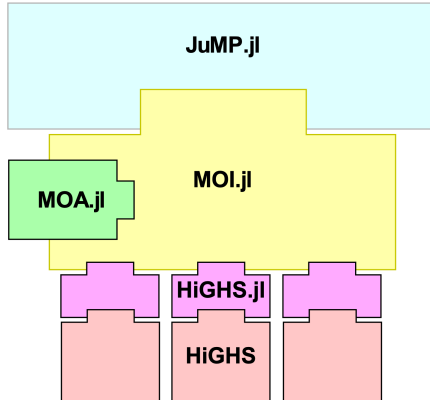
Installing the package:

```
julia> using Pkg
```

```
julia> Pkg.status()  
Status `~/ .julia/environments/v1.11/Project.toml`  
 [60bf3e95] HiGHS v1.18.1  
 [4076af6c] JuMP v1.26.0
```

```
julia> Pkg.add("MultiObjectiveAlgorithms")
```

Getting Started (con't)



Algorithms currently available (v1.11.0)

- | | |
|----------------------------------|------------|
| 1. MOA.Lexicographic() [default] | $p \geq 2$ |
| 2. MOA.Dichotomy() | $p = 2$ |
| 3. MOA.EpsilonConstraint() | $p = 2$ |
| 4. MOA.Hierarchical() | $p \geq 2$ |
| 5. MOA.Chalmet() | $p = 2$ |
| 6. MOA.KirlikSayin() | $p \geq 2$ |
| 7. MOA.DominguezRios() | $p \geq 2$ |
| 8. MOA.TambyVanderpooten() | $p \geq 2$ |
| 9. MOA.GeneralDichotomy() | $p \geq 2$ |

+ optimization attributes coming with a given algorithm

Algorithms currently available (v1.11.0)

- | | |
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+ optimization attributes coming with a given algorithm

Zoom on the ϵ -constraint algorithm

Algorithm:

► `EpsilonConstraint()`

Y.V. Haimes, L.S. Lasdon, D.A. Wismer (1971). On a bicriterion formation of the problems of integrated system identification and system optimization. *IEEE Transactions on Systems, Man and Cybernetics*, Volume SMC-1, Issue 3, Pages 296-297.

Attributes:

► `MOA.EpsilonConstraintStep()`

algorithm uses this value as the ϵ by which it partitions the first-objective's space. The default is 1, so that for a pure integer program this algorithm will enumerate Y_N .

► `MOA.SolutionLimit()`

if this attribute is set then, instead of using the `MOA.EpsilonConstraintStep`, with a slight abuse of notation, `EpsilonConstraint` divides the range of the first-objective's domain in objective space by `SolutionLimit` to obtain the ϵ to use when iterating. Thus, at most `SolutionLimit` solutions are returned.

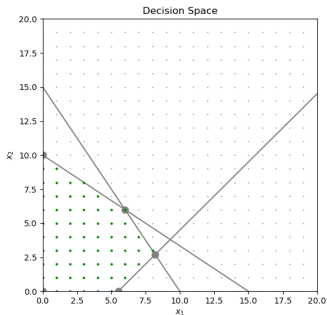
Example (1/4)



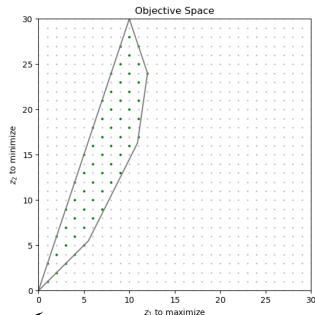
$$\begin{aligned}\max z_1 &= x_1 + x_2 \\ \min z_2 &= x_1 + 3x_2 \\ \text{s.t.} \quad 2x_1 + 3x_2 &\leq 30 \\ 3x_1 + 2x_2 &\leq 30 \\ x_1 - x_2 &\leq 5.5 \\ x_1, x_2 &\in \mathbb{N}\end{aligned}$$

```
julia> c1 = [1,1]
julia> c2 = [1,3]
julia> A = [2 3; 3 2; 1 -1]
julia> b = [30, 30, 5.5]
```

Decision space:



Objective space:





Coding the MOO problem with JuMP

```
julia> using JuMP, HiGHS
julia> import MultiObjectiveAlgorithms as MOA
```

```
julia> model = Model( )
julia> @variable(model, x1 ≥ 0, Int)
julia> @variable(model, x2 ≥ 0, Int)
julia> @expression(model, fct1, x1 + x2)           # to maximize
julia> @expression(model, fct2, x1 + 3 * x2)      # to minimize
julia> @objective(model, Max, [fct1, (-1) * fct2])
julia> @constraint(model, 2*x1 + 3*x2 ≤ 30)
julia> @constraint(model, 3*x1 + 2*x2 ≤ 30)
julia> @constraint(model, x1 - x2 ≤ 5.5)
```



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julia> @constraint(model, 2*x1 + 3*x2 ≤ 30)
julia> @constraint(model, 3*x1 + 2*x2 ≤ 30)
julia> @constraint(model, x1 - x2 ≤ 5.5)
```



Setup the MIP solver: e.g. HiGHS

```
julia> set_optimizer(model, ()->MOA.Optimizer(HiGHS.Optimizer))
```

Setup the algorithm: ϵ -constraint; step=1 (default value)

```
julia> set_attribute(model, MOA.Algorithm(), MOA.EpsilonConstraint())
```

Optimize the MOO problem

```
julia> optimize!(model)
```



Setup the MIP solver: e.g. HiGHS

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Optimize the MOO problem

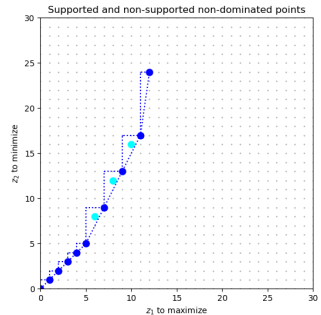
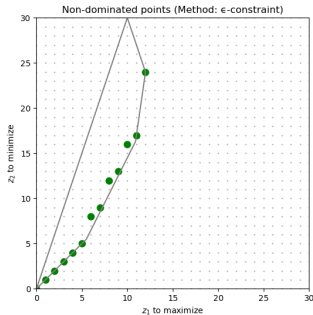
```
julia> optimize!(model)
```

Example (4/4)



Get X_E and Y_N

```
julia> for i in 1:result_count(model)
julia>     z1_opt = objective_value(model; result = i)[1]
julia>     z2_opt = -1 * objective_value(model; result = i)[2]
julia>     x1_opt = value(x1; result = i)
julia>     x2_opt = value(x2; result = i)
julia> end
```



Display of Y_N as a figure

```
julia> using Plots
julia> using LaTeXStrings
julia>
julia> Z1opt = [ ]; Z2opt = [ ]
julia> for i in 1:result_count(model)
julia>     push!(Z1opt, objective_value(model; result = i)[1])
julia>     push!(Z2opt, -1 * objective_value(model; result = i)[2])
julia> end
julia>
julia> plot(Z1opt, Z2opt,
julia>     title = "Nondominated points (e-constraint method)",
julia>     xlabel = L"$z_1$ to maximize",
julia>     xlabel = L"$z_2$ to minimize",
julia>     xlims = (0, 30),
julia>     ylims = (0, 30),
julia>     seriestype = :scatter,
julia>     marker = :circle,
julia>     markersize = 5,
julia>     color = :green,
julia>     legend = false
julia> )
```

Summary (explicit model)



↳ https://github.com/xgandibleux/EURO_MCDM2026/blob/main/3.examples/2IPexplicit.ipynb

From
an explicit to an **implicit** MOO model
with JuMP and MOA

Example (1/3)



$$\begin{aligned}
 \max z_1 &= x_1 + x_2 \\
 \min z_2 &= x_1 + 3x_2 \\
 \text{s.t.} \quad &2x_1 + 3x_2 \leq 30 \\
 &3x_1 + 2x_2 \leq 30 \\
 &x_1 - x_2 \leq 5.5 \\
 &x_1, x_2 \in \mathbb{N}
 \end{aligned}$$

$$\begin{aligned}
 \max z_k &= \sum_{j=1}^n c_j^k x_j & k = 1, \dots, p \\
 \text{s.t.} \quad &\sum_{j=1}^n a_{ij} x_j \leq b_i & i = 1, \dots, m \\
 &x_j \in \mathbb{N} & j = 1, \dots, n
 \end{aligned}$$

```

julia> c1 = [1,1]; c2 = -1 * [1,3]; c = vcat(c1',c2')
julia> a = [2 3; 3 2; 1 -1]
julia> b = [30, 30, 5.5]
julia> m,n = size(a)
julia> p = size(c,1)
    
```

```

julia> md = Model( )
julia> @variable(md, x[1:n] ≥ 0, Int)
julia> @expression(md, z[k=1:p], sum(c[k,j]*x[j] for j=1:n))
julia> @constraint(md, Max, [x[D] for k=1:p])
julia> @constraint(md, [x[D]=b[i], sum([D,j]=x[j] for j=1:n) ≤ b[i]]
    
```

Example (1/3)



$$\begin{aligned}\max z_1 &= x_1 + x_2 \\ \min z_2 &= x_1 + 3x_2 \\ \text{s.t.} \quad 2x_1 + 3x_2 &\leq 30 \\ 3x_1 + 2x_2 &\leq 30 \\ x_1 - x_2 &\leq 5.5 \\ x_1, x_2 &\in \mathbb{N}\end{aligned}$$

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```

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julia> md = Model( )
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julia> @expression(md, z[k=1:p], sum(c[k,j]*x[j] for j=1:n))
julia> @objective(md, Max, [z[k] for k=1:p])
julia> @constraint(md, ct[i=1:m], sum(a[i,j]*x[j] for j=1:n) ≤ b[i])
```

Example (1/3)



$$\begin{aligned}
 \max z_1 &= x_1 + x_2 \\
 \min z_2 &= x_1 + 3x_2 \\
 \text{s.t.} \quad &2x_1 + 3x_2 \leq 30 \\
 &3x_1 + 2x_2 \leq 30 \\
 &x_1 - x_2 \leq 5.5 \\
 &x_1, x_2 \in \mathbb{N}
 \end{aligned}$$

$$\begin{aligned}
 \max z_k &= \sum_{j=1}^n c_j^k x_j & k = 1, \dots, p \\
 \text{s.t.} \quad &\sum_{j=1}^n a_{ij} x_j \leq b_i & i = 1, \dots, m \\
 &x_j \in \mathbb{N} & j = 1, \dots, n
 \end{aligned}$$

```

julia> c1 = [1,1]; c2 = -1 * [1,3]; c = vcat(c1',c2')
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julia> b = [30, 30, 5.5]
julia> m,n = size(a)
julia> p = size(c,1)
    
```

```

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julia> @constraint(md, ct[i=1:m], sum(a[i,j]*x[j] for j=1:n) ≤ b[i])
    
```

Example (1/3)



$$\begin{array}{llll} \max z_1 & = & x_1 & + & x_2 \\ \min z_2 & = & x_1 & + & 3x_2 \\ \text{s.t.} & & 2x_1 & + & 3x_2 \leq 30 \\ & & 3x_1 & + & 2x_2 \leq 30 \\ & & x_1 & - & x_2 \leq 5.5 \\ & & x_1 & , & x_2 \in \mathbb{N} \end{array} \quad \rightarrow \quad \begin{array}{llll} \max z_k & = & \sum_{j=1}^n c_j^k x_j & & k = 1, \dots, p \\ \text{s.t.} & & \sum_{j=1}^n a_{ij} x_j \leq b_i & & i = 1, \dots, m \\ & & x_j \in \mathbb{N} & & j = 1, \dots, n \end{array}$$

```
julia> c1 = [1,1]; c2 = -1 * [1,3]; c = vcat(c1',c2')
julia> a = [2 3; 3 2; 1 -1]
julia> b = [30, 30, 5.5]
julia> m,n = size(a)
julia> p = size(c,1)
```

```
julia> md = Model( )
julia> @variable(md, x[1:n] ≥ 0, Int)
julia> @expression(md, z[k=1:p], sum(c[k,j]*x[j] for j=1:n))
julia> @objective(md, Max, [z[k] for k=1:p])
julia> @constraint(md, ct[i=1:m], sum(a[i,j]*x[j] for j=1:n) ≤ b[i])
```



Setup and optimize

```
julia> set_optimizer(model, ()->MOA.Optimizer(HiGHS.Optimizer))
julia> set_attribute(model, MOA.Algorithm(), MOA.EpsilonConstraint())
julia> optimize!(model)
```

Get X_E and Y_N

```
julia> for i in 1:result_count(model)
julia>     z_opt = [ objective_value(md; result = i)[k] for k=1:p ]
julia>     x_opt = value.(x; result = i)
julia> end
```



Plot Y_N

```
julia> using Plots
julia> Plots.scatter(
julia> [value(z[1]; result = i) for i in 1:result_count(md)],
julia> [-1 * value(z[2]; result = i) for i in 1:result_count(md)];
julia> xlabel = "objective 1",
julia> ylabel = "objective 2",
julia> title = "objective space",
julia> legend = false,
julia> xlims = (0,25),
julia> ylims = (0,25),
julia> aspect_ratio=:equal,
julia> )
```

↳ https://github.com/xgandibleux/EURO_MCDM2026/blob/main/3.examples/2IPim.ipynb

Preliminary Conclusion

Key Takeaways...

You are now able to...

- ▶ identify existing software for multi-objective optimization
- ▶ formulate a multi-objective (linear) optimization model
- ▶ implement and solve it in *Julia* using *JuMP* and *MOA*
- ▶ extract and interpret X_E and Y_N produced by the solver
- ▶ choose an appropriate solution algorithm for a given problem

... use *Julia*, *JuMP* and *MOA* as a integrated workflow for multi-objective optimization

Next Steps...

Going further with JuMP:

- ▶ modify the model data, the model itself
- ▶ advanced modeling techniques, including sets and indices
- ▶ advanced resolution techniques, including callbacks to solvers
- ▶ other classes of optimization problems

Going further with MOA:

- ▶ compare the MOA algorithms with your own algorithm
- ▶ contribute by integrating your own algorithm into MOA

Going further with Julia:

- ▶ discover and learn to code in Julia
- ▶ use packages from among the 12,000 available packages

Références (1/2)...

JuMP (documentation, resources, community, etc.):

<https://jump.dev/>

<https://jump.dev/JuMP.jl/stable/>

Miles Lubin, Oscar Dowson, Joaquim Dias Garcia, Joey Huchette, Benoît Legat, Juan Pablo Vielma. JuMP 1.0: Recent improvements to a modeling language for mathematical optimization. *Mathematical Programming Computation*. 2023.

MultiObjectiveAlgorithms (source code, examples, tutorials):

<https://github.com/jump-dev/MultiObjectiveAlgorithms.jl>

https://jump.dev/JuMP.jl/stable/tutorials/linear/multi_objective_examples/

Oscar Dowson, Xavier Gandibleux, Gökhan Kof. MultiObjectiveAlgorithms.jl: a Julia package for solving multi-objective optimization problems. *INFORMS Journal of Computing*, 2026.

Xavier Gandibleux. Native Multi-Objective Mathematical Programming Software. Chapter 15 in *Handbook on Multi-objective Combinatorial Optimisation* (M. Ehrgott, J. Figueira, Ö. Karsu, L. Paquete Edts.). Springer, 2026 (to appear).

References (2/2)...

Books

Introduction to integer programming and applications with Julia

Luiz Antonio Nogueira Lorena, Luiz Henrique Nogueira Lorena, and Antônio Augusto Chaves
2026

<https://intro-ilp-julia.github.io/book/>

Julia Programming for Operations Research

Changhyun Kwon

2nd Edition. 260 pages. ISBN-10: 1798205475, ISBN-13: 9781798205471.

<https://www.chkwon.net/julia/juliabook/juliabook2.html>

Algorithms for Optimization

Mykel J. Kochenderfer, and Tim A. Wheeler

The MIT Press Cambridge, Massachusetts London, England

<https://algorithmsbook.com/optimization/files/optimization.pdf>